

Title: Quantum Computing Applied to Molecular Modeling

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EXECUTIVE SUMMARY

This project explored the use of quantum computing to calculate the energy of key molecules, H_2 and CO_2 — relevant to green energy and greenhouse gas studies, using first-principles methods. We leveraged the latest computational advancements to assess the method's effectiveness against well-established approaches, such as Full CI. For the calculations, the hybrid classical-quantum Variational Quantum Eigensolver (VQE) algorithm was employed. This algorithm computes the energy spectrum of the electronic structure, which is modeled by Hartree-Fock, by solving the Hamiltonian eigenvalue problem on a quantum computer, and then optimizes the ground-state energy using classical optimizers. The accuracy of the simulated results was assessed using different Hartree-Fock basis sets, varying numbers of shots, and problem reduction techniques such as symmetry analysis and active space selection. The results demonstrated that, with current technology and for the tested molecules, chemical accuracy could be achieved, yielding results comparable to precise classical first-principles calculation methods, such as Full CI.

CONTENTS

1. INTRODUCTION AND MOTIVATION	3
2. METHODOLOGY AND RESEARCH STRATEGY	6
2.1. H ₂ AND CO ₂ ENERGY MODELING TO QUBIT MAPPING	6
2.2. QUANTUM CHEMISTRY LIBRARIES	7
2.3. H ₂ AND CO ₂ ACTIVE SPACE	8
2.4. H ₂ AND CO ₂ ANSATZ	10
2.5. VARIATIONAL QUANTUM EIGENSOLVER (VQE)	12
3. RESULTS, APPLICATIONS AND IMPACT OF WORK	13
3.1. H ₂ MOLECULE	13
3.2. CO ₂ MOLECULE	18
4. CONCLUSION AND FUTURE PERSPECTIVES	19
5. ARTICLES AND CONFERENCES	21
6. REFERENCES	21

1. INTRODUCTION AND MOTIVATION

The increase in CO_2 concentration in the atmosphere has raised alarming concerns about climate change. In response, efforts have emerged to reduce energy sources that emit carbon dioxide by replacing them with green fuels (Noussan *et al.*, 2020). Among these, hydrogen has been extensively researched as a promising candidate for clean and renewable energy, as its only byproduct is water (Yu *et al.*, 2021).

To achieve this, research has been conducted in nanotechnology laboratories, both experimentally and computationally, to evaluate catalysts for the water hydrolysis reaction in hydrogen fuel production. The goal is to maximize energy efficiency and minimize costs, making this technology viable (Dash *et al.*, 2022).

At the same time, there is significant interest in studying reactions involving the CO_2 molecule, as we have already surpassed the point of reversibility regarding the effects of carbon dioxide in the atmosphere. As a result, merely reducing or even halting CO_2 emissions will not solve the problem. Current research is focused on reversing these emissions (Keller *et al.*, 2018). In this context, CO_2 capture projects are being investigated both experimentally and computationally. The primary approach is the mineralization of CO_2 through a carbonation process, aiming to combine carbon-based elements and remove them from the atmosphere (Abraham-A and Tassinari, 2023). Another approach examines the absorption of carbon by the Ni-Hec composite, extensively found in clay formations, potentially leading to the development of efficient greenhouse gas adsorbents (Bø Hunvik *et al.*, 2021).

Reactions involving the synthesis of hydrogen from water and the mineralization of CO_2 can be highly challenging experimentally, with the latter being a process that can take thousands of years in nature (Power *et al.*, 2013). In this context, computational simulation through first-principles calculations is an effective and powerful approach to investigating these processes. This allows reactions to be explored in the presence of various catalysts, in an arbitrary short time and using only computational resources.

First-principles calculations are quite efficient in describing atomic interactions based on the Coulomb force that governs the interaction of subatomic particles, but they are limited to small systems with only a few molecules, as the computational cost grows factorially with the increase in electrons and nuclei. Classical methodologies have successfully achieved accurate models employing the Born-Oppenheimer approximation for nuclei. Regarding the electronic structure, two quantum characteristics are particularly demanding in terms of computational cost: the

indistinguishability of identical particles, yet the antisymmetry of the wavefunction under electron exchange, a consequence of the Pauli exclusion principle, as well as quantum electronic correlations, which have all so far been approximated through electron density-based methods and accounted for in the exchange-correlation term of the successful approach of Density Functional Theory (DFT) (Rahmatpour and Esmaeili, 2024).

This term, which encompasses the approximate modeling of these two quantum characteristics, is far from being exactly computable, even on the most advanced existing supercomputer or any foreseeable future design. This is because emulating quantum characteristics classically, even for small systems, requires space and time resources on the order of the size and age of the universe or even greater.

In this context, quantum computers emerge as a promising approach for effectively modeling quantum effects (Hastings *et al.*, 2014), as these two properties are inherently present in both qubits — quantum bits — and in the nucleus-electron cloud system that we aim to study (McArdle *et al.*, 2020b). As a result, the mathematical overhead associated with classically modeling these effects is eliminated, reducing the problem to the correct choice of ansatz: a properly configured quantum circuit with the necessary correlations to accurately represent the system and enable its study, naturally capturing the intrinsic quantum characteristics of electron exchange and many-body quantum correlations (Smart and Mazziotti, 2020).

The quantum computational resources currently available are still in an early stage, with several challenges to overcome, primarily limited by two main factors: noise and the effect of quantum decoherence, which both reduce the effective computational space (Cai *et al.*, 2023).

Regardless, the current stage and the availability of cloud-based services have enabled the exploration of quantum computing for first-principles calculations of molecules relevant to this project, namely H_2 and CO_2 . The two main limitations when encountered were appropriately addressed and/or mitigated through methodological choices to fulfill the project's objective. Nevertheless, by exploring quantum computing, we are building the knowledge necessary to lay the technological groundwork for significantly advancing the frontiers of nanotechnology and materials science.

2. METHODOLOGY AND RESEARCH STRATEGY

2.1. H₂ and CO₂ energy modeling to qubit mapping

The modeling of interatomic interactions, considering the subatomic composition of atoms, can be described by taking into account the kinetic and interaction energy terms under a Coulomb potential between the particles (nuclei and electrons). The general formulation in the non-relativistic formalism is as follows, where the indices i, j represent the electrons, while A, B represent the nuclei. The momentum of electron i is denoted as p_i , and the momentum of nucleus A is P_A . The masses of the electron and nucleus A are represented by m_e and M_A , respectively. The position of electron i is given by $\mathbf{r}_i$, and the position of nucleus A is $\mathbf{R}_A$. The charge of nucleus A is $Z_A e$, where e is the elementary charge.

$$H = \sum_i \frac{p_i^2}{2m_e} + \sum_A \frac{P_A^2}{2M_A} - \sum_{i,A} \frac{Z_A e^2}{|\mathbf{r}_i - \mathbf{R}_A|} + \sum_{i<j} \frac{e^2}{|\mathbf{r}_i - \mathbf{r}_j|} + \sum_{A<B} \frac{Z_A Z_B e^2}{|\mathbf{R}_A - \mathbf{R}_B|}$$

The simulation of this observable, including both nuclear and electronic degrees of freedom, is still unfeasible on current quantum computers. It would require an extremely large number of qubits, leading to precision loss due to noise, as well as deep circuits, further exacerbating precision loss due to quantum decoherence. As a result, the simulation would yield an irrecoverable outcome due to the high proportion of noise and information loss affecting the data of interest.

The Born-Oppenheimer approximation is a fundamental technique in this context, justified by the order-of-magnitude difference between the masses of nuclei and electrons, which is reflected in the disparity between their velocities. The nuclear kinetic energy term is considered constant in the simulation and incorporated into the energy term. That is, only electron-electron interactions are considered; electron-nucleus interactions are neglected, thus, we assume that electrons instantaneously adjust to the positions of the nuclei due to their higher speed and lower mass, and nucleus-nucleus interactions are treated algorithmically. After applying this approximation, we obtain the following electronic Hamiltonian:

$$H_{\text{BO}} = \sum_i \frac{p_i^2}{2m_e} - \sum_{i,A} \frac{Z_A e^2}{|\mathbf{r}_i - \mathbf{R}_A|} + \sum_{i<j} \frac{e^2}{|\mathbf{r}_i - \mathbf{r}_j|} + \sum_{A<B} \frac{Z_A Z_B e^2}{|\mathbf{R}_A - \mathbf{R}_B|}$$

By adapting this Hamiltonian to the mathematical representation of a two-level system for quantum computing, in terms of the fermionic creation and annihilation operators, we obtain a

Hamiltonian that is isospectral to the electronic Hamiltonian, given by the following equation, where h_{pq} and h_{pqrs} are the one-body and two-body integrals (McArdle *et al.*, 2020a):

$$H_{BO} = \sum_{pq} h_{pq} \hat{a}_p^\dagger \hat{a}_q + \frac{1}{2} \sum_{pqrs} h_{pqrs} \hat{a}_p^\dagger \hat{a}_q^\dagger \hat{a}_r \hat{a}_s,$$

$$h_{pq} = \int \phi_p^*(r) \left(\frac{-\nabla_r^2}{m} - \sum_I \frac{Z_I e^2}{|R_I - r|} \right) \phi_q(r) dr, \quad h_{pqrs} = \int \frac{\phi_p^*(r_1) \phi_q^*(r_2) \phi_r(r_2) \phi_s(r_1)}{|r_1 - r_2|} dr_1 dr_2$$

Now, we can use a canonical mapping algorithm, such as Jordan-Wigner, or a reduced mapping (Setia *et al.*, 2020), such as Bravyi-Kitaev, which exploit the molecular symmetry to reduce the number of resulting operators (Tranter *et al.*, 2018), in order to obtain the Hamiltonian in terms of operators in the Pauli matrix basis (I, Z, X, Y), which form a universal basis for the gates used in quantum computing circuits, making it operable on a quantum computer.

In the case of H_2 , which originally requires 4 qubits and a circuit depth of 14, the mapping can be optimized to a lower complexity, using only 1 qubit and a depth of 2, significantly reducing computational space allowing us to tackle more complex systems.

Table 1. Resulting operators for the H_2 Born-Oppenheimer Hamiltonian

Mapper	Operators	Depth	N. Of Qubits
Jordan-Wigner	$IIIZ + IIZI + IZII$ $+ ZIII + IIZZ + IZIZ + ZIIZ$ $+ YYYY + XXYY + YYXX$ $+ XXXX + IZZI + ZIZI + ZZII$	14	4
Bravyi-Kitaev	$IIIZ + IIZZ + IZII$ $+ ZZZI + IIZI + IZIZ + ZZZZ$ $+ ZXIX + IXZX + ZXZX$ $+ IXIX + IZZZ + ZZIZ + ZIZI$	14	4
Tapered	$Z + X$	2	1

2.2. Quantum Chemistry Libraries

For quantum chemistry simulations, different software frameworks provide specialized tools for handling molecular Hamiltonians and variational algorithms. In our study, we will primarily use Qiskit Nature, a module of the Qiskit library developed by IBM for quantum chemistry applications.

Additionally, we will explore PennyLane Chemistry, developed by Xanadu, as well as OpenFermion over Cirq, developed by Google, which offer alternative approaches to quantum chemistry simulations.

Beyond quantum computing frameworks, we incorporate classical electronic structure methods through PySCF, a widely used open-source library for quantum chemistry calculations. PySCF enables high-accuracy simulations and serves as a backend for generating molecular integrals, which are essential for constructing the Ansatz in first-principles methods. The quantum chemistry frameworks explored in this study — Qiskit Nature, OpenFermion over Cirq, and PennyLane Chemistry — utilize PySCF as a classical computation layer to characterize molecular systems before mapping them to quantum circuits.

2.3. H_2 and CO_2 active space

The hydrogen molecule (H_2), as a two-electron system, holds unique theoretical and didactic value due to its structural simplicity. Despite that, it does not have an analytical solution. The work of Kolos and Wolniewicz (1968) established H_2 as a key benchmark for validating quantum chemistry methods. Altogether, its simplicity eliminates the need for problem reduction techniques, such as active space selection, making it an ideal test case for direct computational approaches.

First, it is notable that the carbon dioxide molecule is significantly more complex than our simplest case of H_2 . This complexity arises from the fact that this molecule consists of three atoms, each of which is more intricate than a hydrogen-like atom, meaning they have more electrons in their valence shell.

Moreover, there are additional orbitals involved in bonding beyond the $1s^2$ orbital, which plays a central role in H_2 . In the case of carbon, we have the $1s^2 2s^2 2p^2$ configuration, while oxygen has $1s^2 2s^2 2p^4$. As a result, there are a total of five molecular orbitals ($1s$, $2s$, $2p_x$, $2p_y$, $2p_z$) for each atom. However, we know that in molecular bonding, only certain orbitals play a predominant role in influencing the bonding interactions (Lawley, 2009).

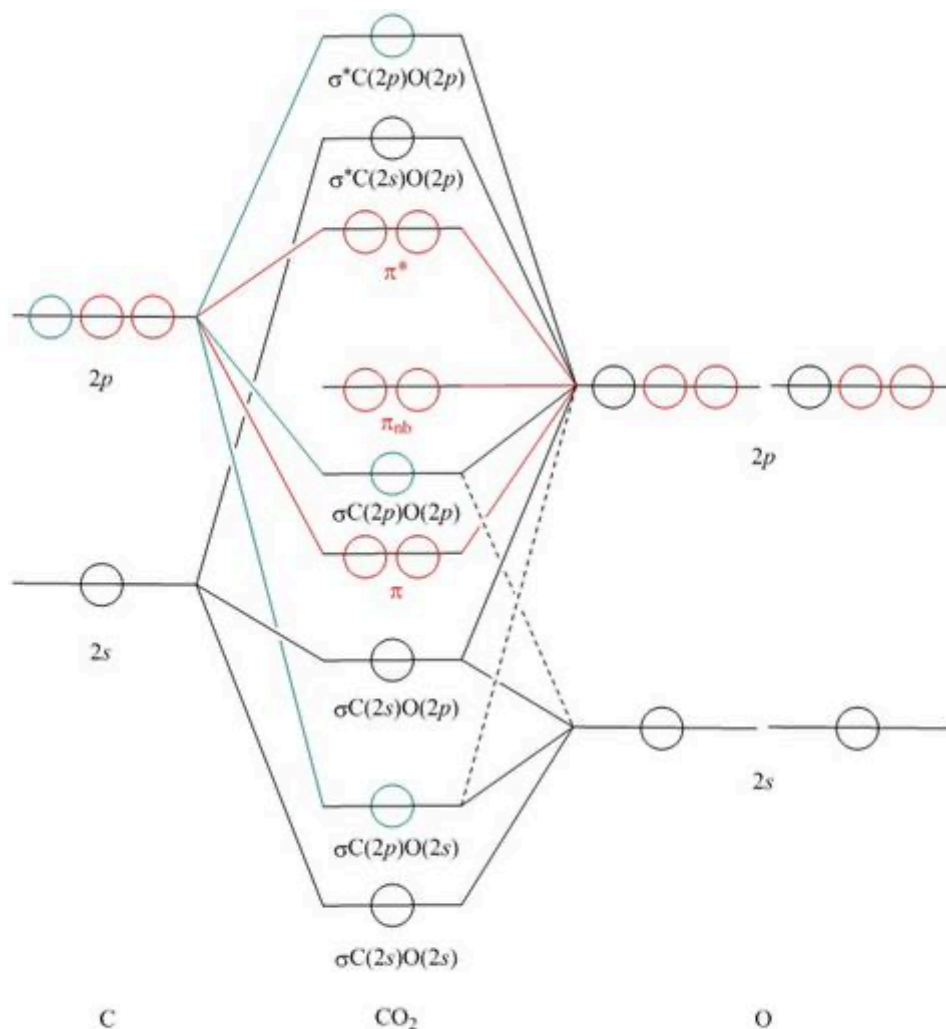


Figure 1. Energy level diagram for the CO_2 molecule.

It is important to note this because when handling the molecule in Qiskit, it considers the molecule as a whole, meaning all electrons and complete molecular orbitals are taken into account. Therefore, we must be mindful of this when dealing with molecules more complex than H_2 .

To address this, we will analyze the case where we use an Active Space, which is a selected set of spatial orbitals used to describe the molecule (Lawley, 2009). We will use active spaces of (8,8) and (4,4), where these numbers correspond to the number of orbitals and electrons, respectively.

When creating the CO_2 molecule, the number of fermionic operators used to describe the Hamiltonian of the generated molecule is:

Table 2: Fermionic operators for the full description of CO₂ orbitals.

Space	N. Of Operators
Complete	44.898
Active Space (8,8)	3.620
Active Space (4,4)	456

Thus, when mapping to the qubit space using the reduced mapping, we obtain the following circuit depths and number of qubits for each case:

Table 3: Depth and number of qubits for each space complexity description.

Space	Depth	N. Of Qubits
Complete	16.170	26
Active Space (8,8)	1.289	12
Active Space (4,4)	116	5

It is notable that by employing appropriate approximations for the problem at hand, we can reduce the complexity level of the quantum circuit, significantly impacting the computational cost of the simulation, reducing the execution time, and mitigating noise effects.

2.4. H₂ and CO₂ Ansatz

Having obtained the expression for the observable H_{BO} of the system as a composition of operators described in the Pauli basis, the next step is to define an ansatz for the wavefunction Ψ , which will be represented by the quantum circuit itself. The Hamiltonian will then be applied as an observable to measure the energy spectrum of the system of interest and subsequently minimized to obtain the ground-state energy (Armaos *et al.*, 2019).

We performed total energy calculations by varying the distance between atoms using the Variational Quantum Eigensolver (VQE) method. First, a driver is used to construct the quantum state of our molecule, where we define some variable parameters. In this case, the quantum state for the hydrogen molecule was generated. In this context, different classical codes can be used to compute the initial solutions of the system's Hamiltonian. For the most part, the PySCF library was used to calculate the initial solutions via Hartree-Fock.

The following code presents the creation of drivers for a given list of distances, where a driver is generated for each value, allowing for an interval analysis. The function returns an `ElectronicStructureProblem` object, which contains the essential information required for the construction of the Ansatz.

The driver used requires, in addition to the molecular structure, a specified basis set to represent the electronic wavefunction of the spin orbitals that compose the Slater determinant in the Hartree–Fock method. Thus, using different basis sets can yield distinct results for the ground-state energy, making it necessary to conduct an analysis to determine the most suitable basis set for this project.

```
1  from qiskit_nature.units import DistanceUnit
2  from qiskit_nature.second_q.drivers import PySCFDriver
3
4
5  def driver(intervalo, base = "sto3g"):
6
7      for a in intervalo:
8
9          driver = PySCFDriver(
10             atom=f"H 0 0 0; H 0 0 {a}",
11             basis=base,
12             charge=0,
13             spin=0,
14             unit=DistanceUnit.ANGSTROM,
15         )
16
17     yield driver.run()
```

Figure 2. Generating molecular structure data using PySCF in Qiskit Nature.

For the CO_2 molecule, the Qiskit Nature library provides modules for ansatz implementation. The ansatz used throughout this study was the Unitary Coupled Cluster Single-Double (UCCSD), which accounts for single and double excitations of the molecule (Ma *et al.*, 2014).

The effectiveness of the method is limited by the quality of the ansatz, as it determines the minimum energy that can be achieved through optimization (Ollitrault *et al.*, 2021). Various approaches have been explored to develop an adaptive ansatz, including the Adaptive Variational Quantum Eigensolver (Adaptive VQE) (Grimsley *et al.*, 2019); however, this topic is beyond the scope of this work.

2.5. Variational Quantum Eigensolver (VQE)

Variational quantum algorithms (VQAs) have emerged as a promising approach for leveraging noisy intermediate-scale quantum (NISQ) devices to solve optimization and quantum chemistry problems (Tilly *et al.*, 2022). This hybrid quantum-classical approach is particularly useful for quantum chemistry simulations, as it enables the efficient exploration of the molecular energy landscape while mitigating the limitations imposed by quantum hardware noise. The most commonly used VQA for quantum chemistry is the Variational Quantum Eigensolver (VQE), where the optimization targets the lowest eigenvalue of a given Hamiltonian, making it suitable for electronic structure problems.

The figure above, taken from Bharti *et al.* (2022) illustrates the workflow of a VQE, which consists of a quantum-classical loop.

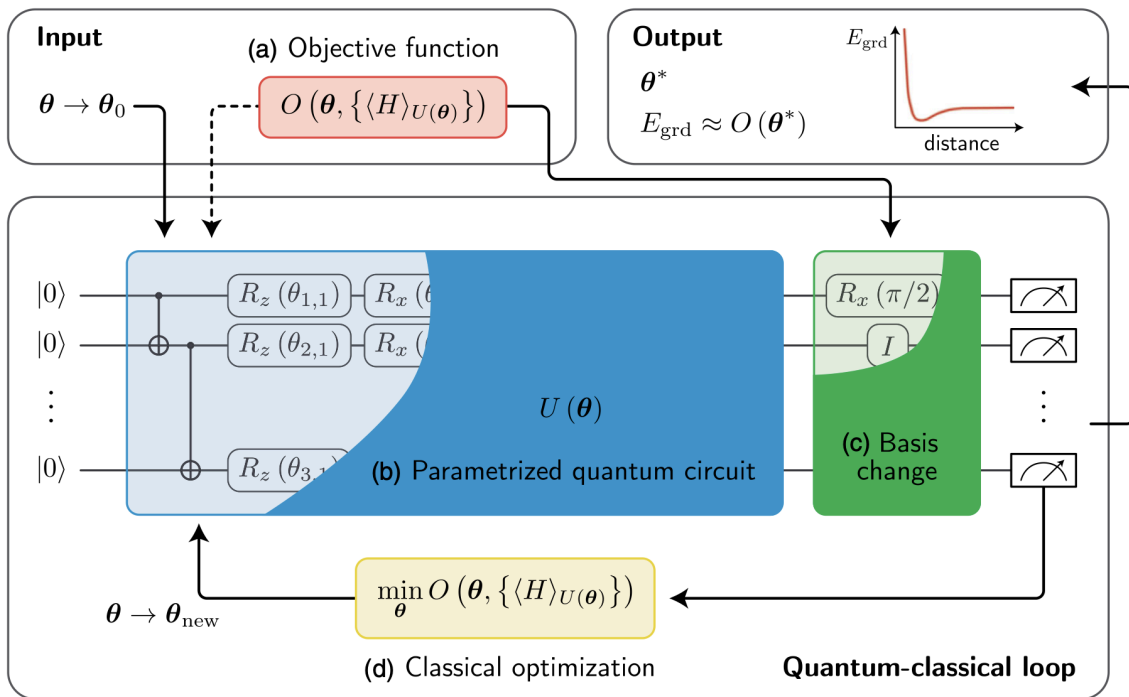


Figure 3. Diagram of the VQE algorithm: qubit initialization, state preparation with ansatz, measurement, and optimization.

In this framework, a parametrized quantum circuit $U(\theta)$ is used to prepare a trial wavefunction, where the parameters θ define the quantum state. The expectation value of the Hamiltonian is then measured, requiring a basis transformation step for different operator terms. This expectation value serves as the objective function $O(\theta)$, which is minimized using a classical optimizer to update the parameters iteratively. The process continues until convergence is reached, yielding an approximation of the system's ground-state energy (Tilly *et al.*, 2022).

3. RESULTS, APPLICATIONS AND IMPACT OF WORK

3.1. H_2 molecule

In Figure 3, we present the energy solution of the Hamiltonian for the H_2 molecule, the nuclear repulsion energy, and the potential energy of the actual molecule. The results show that VQE produces an energy value identical to that of the Full Configuration Interaction (FCI) method, with both being lower than the Hartree-Fock (HF) method and closer to the experimental value.

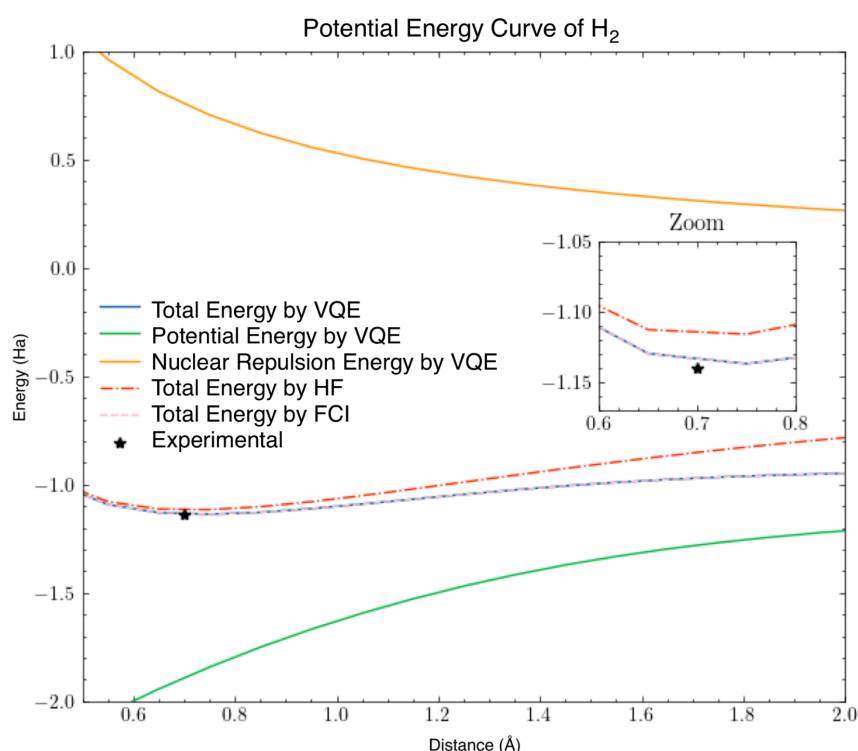


Figure 5. Graph of the potential energy analysis of the H_2 molecule.

The results obtained for the standard simulation of the H_2 molecule across different libraries are as follows. We utilized implementations from Cirq, PennyLane, and Qiskit, with the latter offering greater versatility, more extensive documentation, and easier implementation. All implementations use an Ansatz modeled by the PySCF library, based on Hartree-Fock, with the Gradient Descent algorithm for optimization in VQE.

It can be observed that the energy values obtained using Qiskit and Cirq Full CI are lower and closer to the reference experimental value, given by $E = -1.14$ Ha. In the case of PennyLane, the curve closely follows that of Cirq HF due to the construction of its Ansatz, with slight variations caused by the optimizer used. For these reasons, this library was chosen as the primary tool in our study.

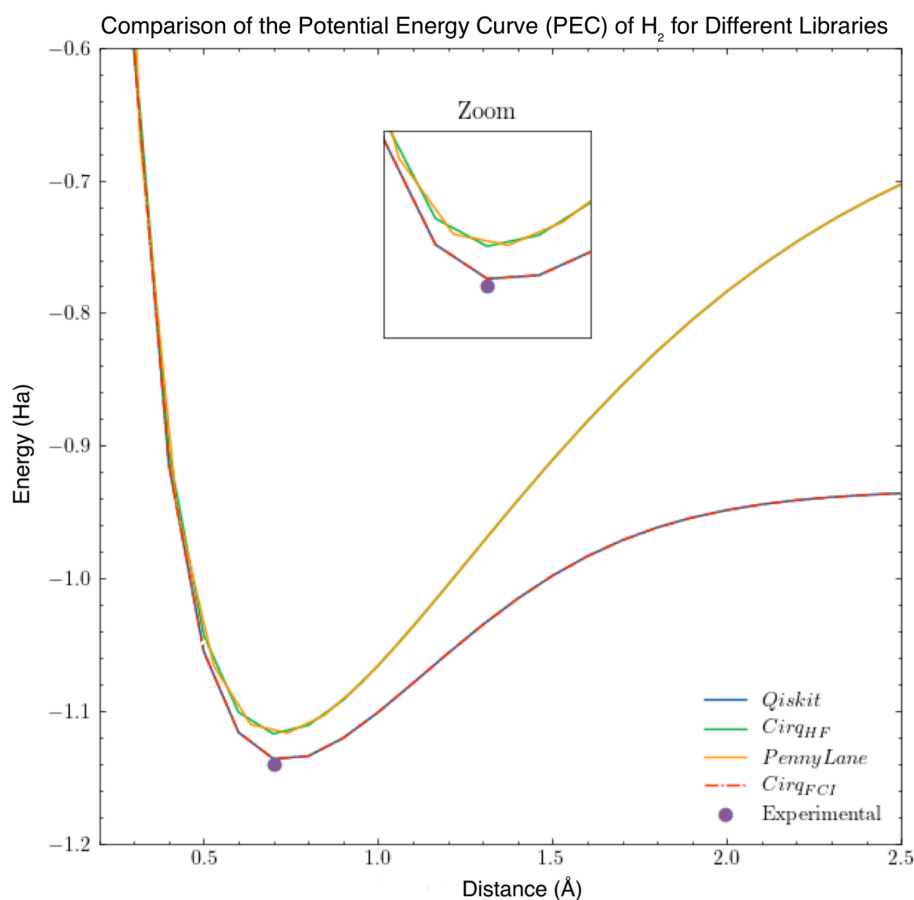


Figure 6. H_2 Simulation Analysis Across Libraries.

Since we are using the VQE approach, which is a variational algorithm, the choice of optimizer plays a crucial role. Therefore, we explored which optimizer best balances solution quality and scalability. We construct the standard H_2 molecular system with the configurations described in the methodology and vary only the optimizer used. This will allowed us to assess which optimizer provides the best trade-off between data accuracy and processing time. The quality of the results obtained can be visualized in the potential energy curve graphs in Figure 5.

We observe that, although most optimizers yield nearly overlapping results, careful consideration is required when selecting an optimizer, especially concerning convergence time, which is directly related to the scalability of the method. If an excessively costly optimizer is used, it may hinder the ability to explore more complex molecules due to scalability limitations in the simulation. Based on the potential energy curve and simulation time, the best optimizer was chosen to balance result quality and computational efficiency. The Sequential Quadratic Programming (SLSQP) and Analytic Quantum Gradient Descent (AQGD) optimizers met the proposed criteria, as expressed in table 4.

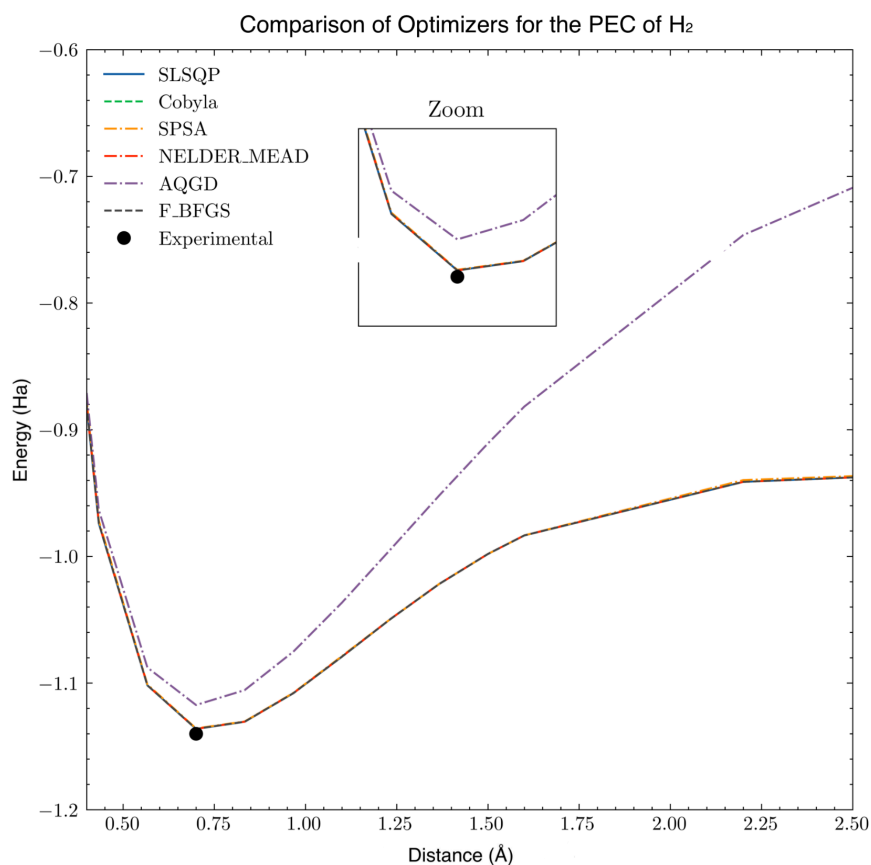


Figure 7. Potential energy curve of H_2 using various optimizers.

Table 4: Optimizer Performance in Potential Energy Curve Computation.

Optimizer	Elapsed Time (s)	Standard Deviation (s)
SLSQP	5.20	0.39
Cobyala	6.59	0.31
SPSA	22.46	1.10
Nelder-Meads	8.22	0.35
AQGD	4.84	0.43
F-BFGS	7.42	0.45

Figure 8 and Table 5 below present the results of the potential energy curve analysis using different basis sets. For the graph, all predefined basis sets available in the PySCF library were tested, and the ten most relevant sets were selected from them.

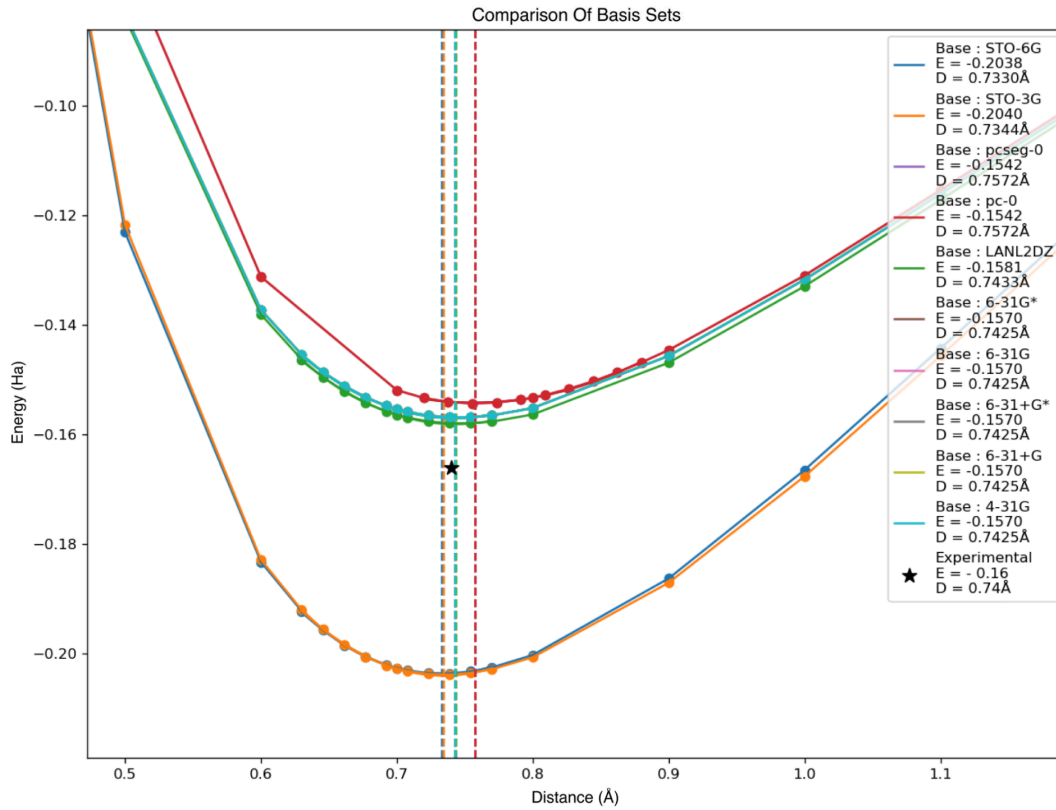


Figure 8. Potential Energy Curve of H_2 for Various Basis Sets.

Table 5: Comparison of Performance and Accuracy of Basis Sets in H_2 Calculations.

Base	Elapsed Time (s)	Equilibrium Distance (Å)
STO-6G	7.35	0.7330
STO-3G	7.57	0.7344
pcseg-0	113.94	0.7572
pc-0	107.47	0.7572
LANL2DZ	110.26	0.7433
6-31G*	106.85	0.7425
6-31G	107.33	0.7425
6-31+G*	106.51	0.7425
6-31+G	106.40	0.7425
4-31G	106.88	0.7425
Experimental	–	0.7400

As observed in Table 5, the results obtained for all basis sets are within 2.5% of the experimentally measured value. However, due to the processing time of each basis set, STO-3G was chosen for all subsequent potential energy curve plots, as it offers a relatively fast processing time and maintains accuracy when comparing equilibrium distances with the experimental measurement.

Due to the probabilistic nature of quantum computing, we observe that as the number of shots — the number of times a single measurement is simulated — increases, the evaluated result for a given point in our simulation becomes more stable. This statistical precision improvement leads to a convergence toward the true expected value, as shown in the Figure 9.

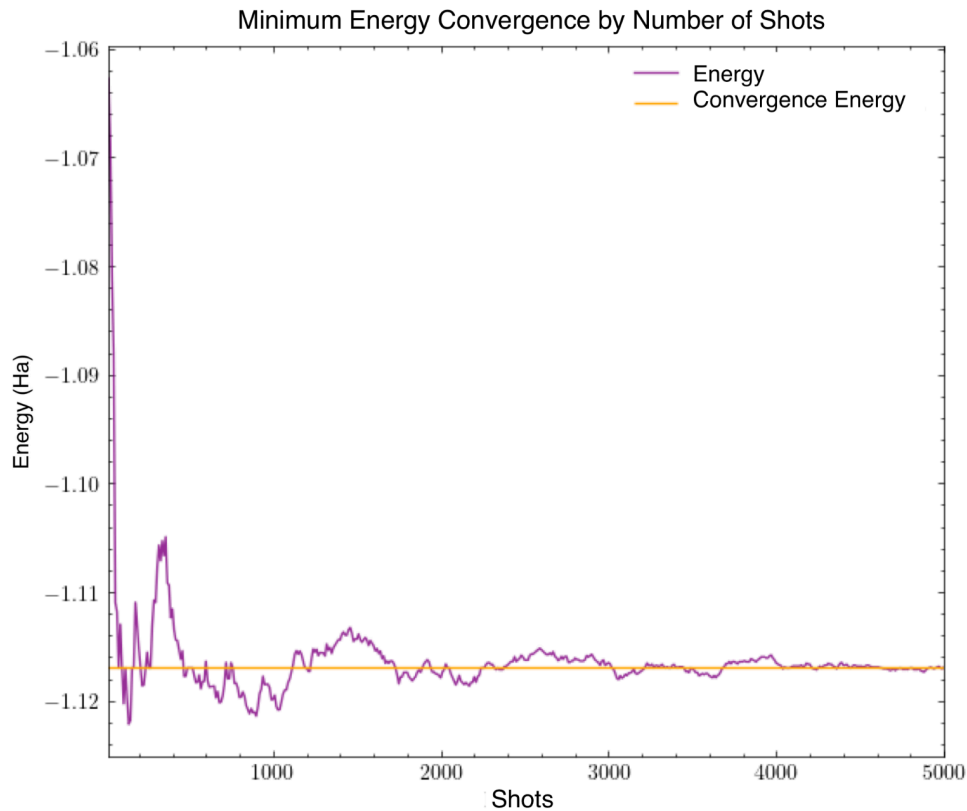


Figure 9. Effect of Shot Count on Minimum Energy Convergence

We observe that around 2000 shots, the energy value stabilizes closer to the expected result. At 4000 shots, the fluctuations are significantly reduced, leading to a much more consistent energy estimation. Thus, beyond 2000 shots, the computed energy values remain close to the convergence value, with minimal deviation.

To better understand the effects of noise, the Qiskit Aer library was employed, enabling noise modeling to simulate actual quantum hardware architectures, thereby obtaining results that better reflect the current technological landscape. To assess the impact of noise, a comparison was made between different noise models based on publicly available architectures from IBM's platform: IBM Kyoto, IBM Osaka, and IBM Brisbane, each a 127-qubit quantum computer. The analysis was performed using 2048 shots.

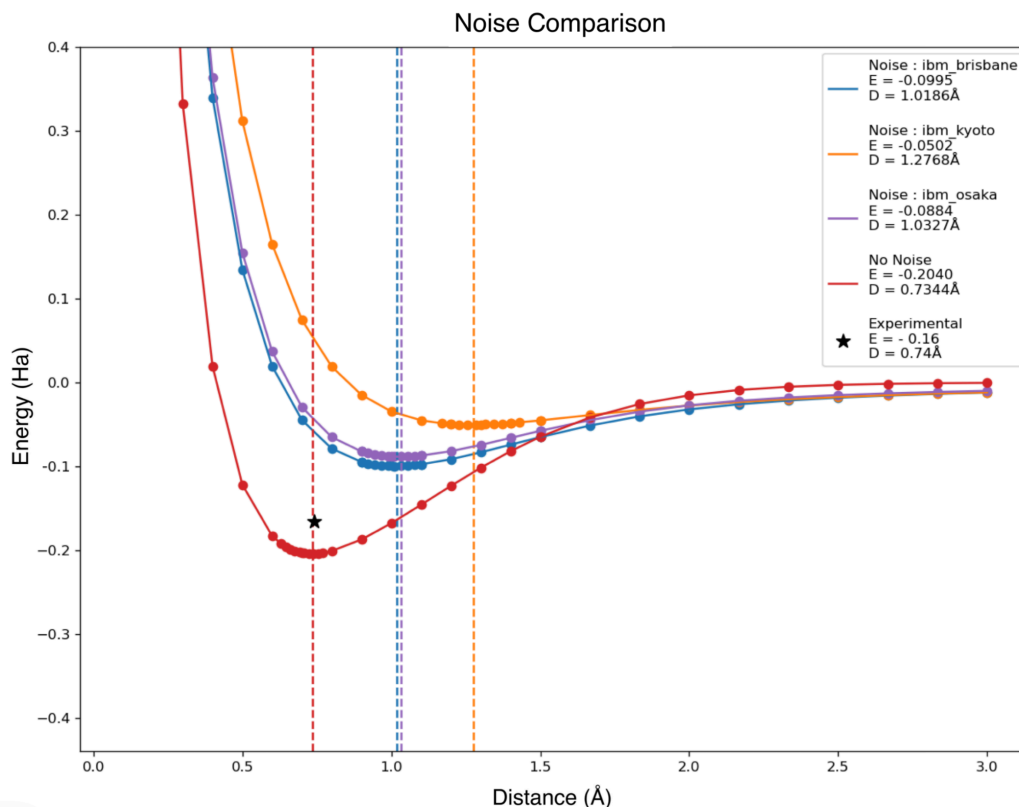


Figure 10. Impact of Noise on Potential Energy Curves for H_2 .

From this curve, we conclude that noise in the simple H_2 circuit does not distort the potential energy curve, although it leads to a loss of chemical accuracy in the problem.

3.2. CO_2 molecule

For the simulation of the CO_2 molecule, we will use the (4,4) Active Space, which considers only the π and π^* bonds, as they are closest to the Fermi level. This approach enables a lower computational cost while potentially introducing deviations from the experimental value.

By employing the (4,4) Active Space method and the reduced mapping, the energy of the molecule was simulated under the assumption that the oxygen atoms remain at an equilibrium distance of 1.16 Å from the carbon atom in CO_2 , yielding a ground-state energy of -186.9240 Ha, with a total simulation time of 54 seconds. Despite the approximations used, the result is comparatively close to other methods (Ma *et al.*, 2014).

However, when using the same configurations but increasing the Active Space to (8,8), the computed energy reaches -186.9623 Ha, but at a much higher computational cost, requiring 76 minutes and 24 seconds to complete the energy calculation.

Notably, using the minimal Active Space introduces an energy deviation of -0.0382 Ha, which is quite significant compared to the chemical accuracy threshold of 1.6×10^{-3} Ha. Nevertheless, this approximation will be adopted to explore the overall energy behavior of the molecule.

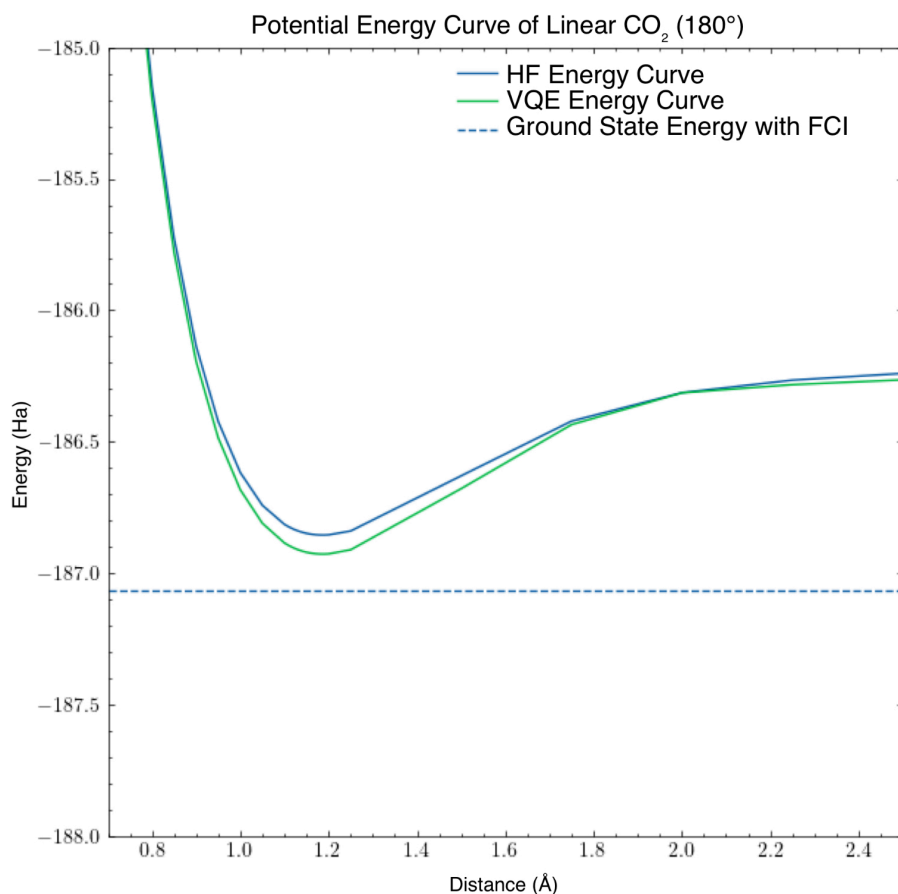


Figure 9. Potential Energy Curve of Linear CO_2 : Comparison of HF, VQE, and FCI Methods.

A potential well behavior can be observed for the ground state of CO_2 , similar to that of H_2 . Additionally, when compared to the Hartree-Fock (HF) method, the obtained ground-state energy is lower, suggesting that VQE is a more accurate approach than HF. However, it remains less precise than Full Configuration Interaction (FCI), which, despite its higher accuracy, comes with a significantly higher computational cost.

4. CONCLUSION AND FUTURE PERSPECTIVES

This project represents a significant milestone in the application of quantum computing to molecular modeling, with a practical example applied to the hydrogen (H_2) and carbon dioxide (CO_2) molecules, both of which play a crucial role in the energy transition. Notable progress was

achieved in understanding electronic states and molecular properties by applying principles of quantum computing.

The application of the Variational Quantum Eigensolver (VQE) algorithm was fundamental for calculating ground-state energies of molecular systems, demonstrating both the feasibility and challenges of this approach. The results indicate that quantum computing has the potential to provide efficient solutions for complex electronic structure problems, even in its early stage, especially when leveraging circuit optimization strategies and qubit mapping techniques.

The CO_2 molecule simulation highlighted the importance of complexity reduction strategies, such as the use of active spaces and qubit-reduced mapping, which significantly reduced computational costs — though at the expense of energy accuracy. Nevertheless, the trade-off between simulation complexity and energy precision remains an important consideration.

The study also underscores the difficulties and limitations of NISQ (Noisy Intermediate-Scale Quantum) devices, which are highly affected by noise, circuit size limitations, and decoherence effects. Additionally, it is important to highlight that the computational cost and execution time reported here were obtained from a classical simulation of quantum computer behavior. This implies that scalability increases with the number of qubits and circuit depth. However, on an actual quantum computer, these computations would be significantly faster, as they naturally align with the intrinsic behavior of quantum hardware — with the caveat that extensive noise mitigation would be required.

Advancements in both quantum algorithms and hardware engineering create vast opportunities for the future development of materials science and molecular simulations, with various fields standing to benefit from these improvements.

The development of specialized human resources in quantum computing and its application to emerging challenges were key pillars, laying a strong foundation for the future of research and technological development in Brazil. The results obtained have the potential to influence and advance the field of renewable energy, fostering cleaner and more sustainable solutions.

This work outlined the process of building foundational knowledge for an initial exploration of quantum computing. As a second step, which is already underway and yielding initial results, this methodology is being applied to actual quantum hardware, with a comparative analysis of error mitigation algorithms, specifically Zero Noise Extrapolation (ZNE) and Clifford Regression. Once an effective error mitigation methodology is established, the range of analyzed systems can be expanded in tandem with the increasing capabilities of quantum hardware.

Thus, the progress described in this work reinforces the promising role of quantum computing in materials science and catalysis research, paving the way for future studies that can explore more complex molecular systems with greater precision and efficiency.

5. ARTICLES AND CONFERENCES

An article is being prepared to publicize the second part of this work, which is currently in progress, as described in the conclusion section.

6. REFERENCES

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